

TravelEase

Travel Assistance With Pooling System

Mini Project Report

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Reg. No.: AJC21MCA-I028

In Partial fulfillment for the Award of the Degree of

INTEGRATED MASTER OF COMPUTER APPLICATIONS

(INMCA)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY



AMAL JYOTHI COLLEGE OF ENGINEERING

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2021-2026

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CERTIFICATE

This is to certify that the Project report titled “**TravelEase**” is the bona fide work of **Binziya P (Regno: AJC21MCA-I028)** carried out in partial fulfillment of the requirements for the award of the **Degree of Integrated Master of Computer Applications** at **Amal Jyothi College of Engineering Autonomous, Kanjirappally**, Affiliated to **APJ Abdul Kalam Technological University**. The project was undertaken during the period from **July 01, 2025** to **October 23, 2025**.

Ms Jetty Benjamin
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Dr. Bijimol T K
Head of the Department

DECLARATION

I hereby declare that the project report “**TravelEase**” is a bona fide work done at **Amal Jyothi College of Engineering Autonomous, Kanjirappally**, Affiliated to **APJ Abdul Kalam Technological University**, towards the partial fulfilment of the requirements for the award of the **Integrated Master of Computer Applications (MCA)** during the period from **July 01, 2025** to **October 23, 2025**.

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ACKNOWLEDGEMENT

First and foremost, I thank God almighty for his eternal love and protection throughout the project. I take this opportunity to express my gratitude to all who helped me in completing this project successfully. It has been said that gratitude is the memory of the heart. I wish to express my sincere gratitude to our Director (Administration) **Rev. Fr. Dr. Roy Abraham Pazhayaparampil** and Principal **Dr. Lillykutty Jacob** for providing good faculty for guidance.

I owe a great depth of gratitude towards our Head of the Department **Dr. Bijimol T K** for helping us. I extend my whole hearted thanks to the project coordinator **Ms Meera Rose Mathew** for her valuable suggestions and for overwhelming concern and guidance from the beginning to the end of the project. I would also express sincere gratitude to my guide **Ms Jetty Benjamin** for her inspiration and helping hand.

I thank our beloved teachers for their cooperation and suggestions that helped me throughout the project. I express my thanks to all my friends and classmates for their interest, dedication, and encouragement shown towards the project. I convey my hearty thanks to my family for the moral support, suggestions, and encouragement to make this venture a success.

BINZIYA P

ABSTRACT

TravelEase Hub is a full-stack web application designed to simplify and personalize travel planning. Unlike existing platforms that offer static listings and lack collaboration features, TravelEase Hub provides a smarter, more personalized experience for travelers, travel agencies, and administrators. Key features include AI-powered itinerary planning, smart package search, booking management, and budget estimation. Future extensions will introduce user pooling and smart expense splitting to handle group travel logistics and cost sharing.

Objectives

The primary goal of TravelEase Hub is to create a comprehensive, user-oriented platform that simplifies travel planning for all users. For travelers, the system aims to deliver a personalized experience with AI-generated itineraries and tools for booking, cancellation, and budget estimation. It will address the complexities of group travel, which existing systems overlook, through future features like matching travelers with similar destinations (user pooling) and automatically dividing costs (smart expense splitting). For travel agencies, the objective is to provide a secure channel to manage listings, create promotions, and receive real-time booking alerts. Administrators will maintain platform integrity by monitoring activity, managing accounts, and verifying submitted travel packages.

Methodology

The TravelEase Hub is being developed as a modern full-stack web application, utilizing a robust and specialized technology stack. The frontend is built using React.js to create interactive and personalized user interfaces. The backend infrastructure relies on Node.js with Express.js to manage application logic and server-side operations. For data management, the system uses Supabase, a PostgreSQL-based backend-as-a-service, which provides a reliable database solution. Core intelligent functionalities, such as the AI Itinerary Planner, will be powered by integration with TensorFlow.js or the OpenAI API. Other essential services include the Google Maps API for displaying travel routes and destinations and JWT (JSON Web Token) for secure user and agency authentication. The entire system is modularly organized into three main components—the User Module, Agency Module, and Admin Module—to clearly define and manage the different sets of features for login/registration, package management, booking, verification, and oversight.

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List of Abbreviations

AI - Artificial Intelligence

API - Application Programming Interface

CSS - Cascading Style Sheets

DB - Database

HTTP - Hypertext Transfer Protocol

JS - JavaScript

JSON - JavaScript Object Notation

JWT - JSON Web Token

UI - User Interface

URL - Uniform Resource Locator

UX - User Experience

React - Frontend framework

Node.js - Backend runtime

Express.js - Web framework

Supabase - Database platform

PostgreSQL - Database system

Vite - Build tool

CHAPTER 1

INTRODUCTION

1.1 PROJECT OVERVIEW

TravelEase Hub is a comprehensive full-stack travel booking platform connecting travelers with agencies through an innovative package management system. Built with React 19.1.1 frontend and Node.js/Express backend, it features three user roles: travelers browsing packages, agencies creating packages (pending admin approval), and administrators managing the platform. The system includes Google OAuth authentication, OTP email verification, and a sophisticated 9-step package creation wizard with real-time accommodation suggestions, restaurant recommendations, and itinerary planning. Leveraging Supabase for database management and authentication, it incorporates intelligent fallback mechanisms, caching strategies, and OpenStreetMap integration for route optimization. Security is ensured through bcrypt hashing, JWT tokens, rate limiting, and Row Level Security policies. The platform features responsive design with SCSS modules, dark mode support, and AI-powered itinerary generation, making it a production-ready solution for real-world travel booking demands.

1.2 PROJECT SPECIFICATION

TravelEase Hub is a cloud-based travel booking platform that enables seamless collaboration between travelers and travel agencies through an innovative package management ecosystem. The platform serves as a unified system allowing users to browse, book, and manage travel packages while providing agencies with comprehensive tools to create detailed travel experiences with real-time accommodation suggestions, restaurant recommendations, and itinerary planning. Advanced geocoding and mapping integration through Google Places API and OpenStreetMap enables precise destination selection and route optimization, while intelligent fallback mechanisms ensure 100% platform uptime. The system operates on a sophisticated three-tier user architecture encompassing travelers seeking packages, agencies creating packages (subject to administrative approval), and administrators managing platform operations through comprehensive approval workflows. Built with React 19.1.1 frontend and Node.js/Express backend infrastructure, the platform integrates Supabase for database management and authentication, implementing Google OAuth 2.0, OTP email verification, and JWT token security protocols. Real-time pricing calculations, AI-powered itinerary generation, and responsive design with SCSS modules make TravelEase Hub a production-ready solution that transforms the entire travel booking experience through seamless user interfaces and automated backend processing.

CHAPTER 2

SYSTEM STUDY

2.1 INTRODUCTION

TravelEase Hub is a full-stack web application designed to simplify and personalize travel planning. The platform serves three types of users—travelers, travel agencies, and administrators—and offers features such as AI-powered itinerary planning, smart package search, booking management, budget estimation, and upcoming tools like user pooling and automatic expense splitting. With interactive dashboards and real-time updates, the platform aims to provide a smarter, more collaborative, and highly personalized travel experience.

2.2 LITERATURE REVIEW

A broad literature exists across recommender systems, group decision-making, digital tourism, and expense-management. Recommender system surveys lay out collaborative, content based, and hybrid approaches that underpin modern personalization engines [1], [2]. In travel-specific contexts, itinerary recommendation and POI (point-of-interest) suggestion literature has advanced using matrix factorization, sequential modeling, and deep learning to capture temporal and contextual preferences [3]–[5]. Group travel and collaborative decision-making have been explored in the context of social choice and multi-stakeholder recommendation; solutions often aim to reconcile heterogeneous preferences using aggregation strategies or fair-share optimization [6], [7]. Pooling and ride-sharing literature provides transferable methods for matching users based on spatio-temporal similarity and cost-optimization [8], [9]. Expense splitting has been addressed by several commercial and academic works that propose algorithms for minimal transaction settlement, fairness criteria, and privacy-preserving ledgering [10], [11]. Research on blockchain-backed verification in travel has proposed immutable receipts and tamper-evident vaccination or ticket records, though practical adoption remains limited [12], [13]. Tourism research highlights the trend toward smart tourism ecosystems, integrating IoT, AI, and context-aware services to create seamless user experiences [14]. Personalization is further enhanced by using predictive analytics and social recommendation methods [15]–[17]. Despite these advances, current commercial applications rarely combine itinerary personalization with collaborative features such as pooling and automatic expense reconciliation. This paper synthesizes these lines of work into a single integrated platform for individual and group travel. Compared to existing travel apps (MakeMyTrip, Expedia, Booking.com), TravelEase Hub emphasizes collaborative planning and automated financial reconciliation, bringing research ideas into a cohesive, user-centered application.

2.3 PROPOSED SYSTEM

The proposed TravelEase Hub platform allows users to register, search for travel packages using smart filters, and receive AI-generated itinerary suggestions based on personal preferences. Users can book and cancel trips, estimate budgets, view travel routes on maps, and access trip history through a personalized dashboard. Travel agencies can sign up to list and manage travel packages, create promotional offers, and receive real-time booking alerts. Administrators are responsible for monitoring platform activity, managing user and agency accounts, and verifying the authenticity of submitted packages. Future extensions of the system will include a user pooling feature that matches travelers with similar destinations and a smart expense splitting tool to automatically divide trip costs among group travelers.

2.4 ADVANTAGES OF PROPOSED SYSTEM

- Highly Personalized Itinerary Planning
- Automated Financial Management
- Comprehensive Trip Management for Travelers
- Transparent Route Visualization
- Smarter Package Search and Filtering.
- User-Friendly Interface

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 FEASIBILITY STUDY

Feasibility is defined as the practical extent to which a project can be performed successfully. A feasibility study determines whether the solution proposed for accomplishing the requirements is practical and workable. It includes considerations such as resource availability, development costs, expected benefits, and maintenance expenses. The outcome should guide whether to proceed with requirements engineering and system development. If a system does not support the business objectives, it holds no real value to the organization. Many systems fail due to unclear goals, undefined requirements, or organizational issues influencing procurement. Therefore, the objective of this feasibility study is to ensure that TravelEase Hub is acceptable to users, adaptable to change, and aligned with recognized standards.

3.1.1 Economical Feasibility

The proposed TravelEase Hub project is determined to be economically feasible. Economic feasibility assesses whether the software will generate financial gains, considering costs such as those incurred on the software development team, hardware and software acquisition, and the feasibility study itself. The project addresses these costs strategically by utilizing open-source technologies, which significantly reduces licensing and infrastructure expenses. By relying on cloud-based deployment, the platform eliminates the need for expensive physical hardware. Revenue will be generated through bookings and optional agency registrations. The platform supports its economic model by offering essential features for free, which increases accessibility and encourages a broader user base. This combination of low development costs and high user value ensures sustainability. As the user base expands and engagement increases, the system is expected to produce long-term revenue through expanded services, ultimately making the project financially beneficial for its stakeholders.

3.1.2 Technical Feasibility

The technical feasibility of the TravelEase Hub project is strongly established by assessing the required current resources and technology needed to accomplish user requirements within the allocated time and budget. The project is built using a stable and widely adopted technology stack, ensuring successful development. The core stack includes React.js for a dynamic and responsive frontend, and Node.js with Express.js for a scalable backend system. For data management, it relies on Supabase, a PostgreSQL-based backend-as-a-service. The implementation of intelligent features like itinerary planning and budget estimation is achieved

through the integration of AI and ML technologies such as TensorFlow.js and the OpenAI API. Furthermore, the system incorporates the Google Maps API for map integration and JWT for secure user authentication. Deployment is efficiently managed using Vercel for fast, serverless hosting. These strategic technology choices confirm that the system can be developed efficiently using established tools with strong community support, thus satisfying the criteria for technical feasibility.

3.1.3 Behavioral Feasibility

Behavioral feasibility, often closely related to operational feasibility, focuses on how the TravelEase Hub project will be received by and integrated into the routines of its users and the organization. The core of this project is highly aligned with modern user behavior, as it directly addresses the dissatisfaction with current travel platforms lacking personalization and collaborative features. The system is designed for quick user adoption due to its user-friendly features, including a clean UI/UX, and specialized dashboards for both users and agencies which minimize the learning curve. The implementation of AI-based suggestions and real-time updates caters to the expectation of personalized, dynamic services. Furthermore, the introduction of group coordination tools like pooling and expense splitting directly solves high-priority user problems often anticipated in group travel, ensuring the suggested solution is acceptable and practical. Travel agencies are also catered to with simplified package management and real-time booking alerts, ensuring they too can quickly integrate the system into their existing operations. Because the platform aligns with current user behavior and offers solutions to highly prioritized problems, both users and agencies are expected to adapt quickly and seamlessly integrate the system into their daily operations.

3.1.2 Feasibility Study Questionnaire

- What are the current challenges you face in finding compatible travel partners and coordinating travel arrangements?

The gradual method of partner selection that depends on matching users according to similar interests together with their past travel records requires extensive manual work.

- How do you currently search for and book tour packages, and what improvements would you like to see in this process?

The user now understands inefficiency when they need to search for tour packages on various platforms, since they have to manually compare prices. The user gets frustrated since the system only gives them generic packages. The system also fails to incorporate

features like pairing partners into the booking.

- What features would you find most useful in a tour package booking system?

The most important functionality includes trip package recommendations and matching platform tools for partners that have a user-friendly booking interface.

- What are the common difficulties that users face while planning group trips? Splitting and managing group bills is usually done manually, and platforms rarely offer built-in tools to simplify this process. Would users prefer AI-generated itineraries or manual customization?

They usually like AI-generated itineraries valuable because it saves them time and provides a good starting point.

- Are users interested in pooling travel with others to reduce costs?

Absolutely, many users, especially solo travellers, students, or selective travellers are very interested in pooling features. Pooling enables travel to a shared destination to be shared by travellers in terms of transportation, accommodation, or activity costs, making it more affordable and on occasion more enjoyable through socialization.

- What are the top filters users prefer while searching for travel packages?

When searching for travel packages, users generally filter by destination, budget, duration of the trip, and travel dates.

- In your opinion, how can system notifications improve the experience for both your agency and your customers?

Definitely, system notifications can make a big difference for us. If we receive immediate alerts when a customer makes a booking or sends an inquiry, we can reply quicker, which is trust-building and enhances our service. Sometimes, delays in responding can cause us to lose potential customers. Also, if the customer receives timely updates about their booking status, payment reminders, or changes in travel plans, it avoids a lot of confusion.

- How do you currently receive and manage customer bookings? Is it manual or digital?

Currently, the majority of our reservations are made over the phone, via WhatsApp, or through social media requests. A digital system with automated booking logs and reminders would really help us avoid errors and save time. Do you currently use any other digital platforms or tools to manage your travel business? Not really. We mostly

depend on WhatsApp, Google Sheets, and some basic email templates.

3.1.5 Geotagged Photograph



3.1 SYSTEM SPECIFICATION

3.2.1 Hardware Specification

Processor - Intel Core i5

RAM - 16GB

Hard disk - 256 GB SSD

3.2.2 Software Specification

Front End - React.js

Back End - Node.js ,Express.js

Database - Supabase

Client on PC - Windows 7 and above.

Technologies used - JS, CSS, Direct License API, OpenAI API

3.3 SOFTWARE DESCRIPTION

3.3.1 Eg.PHP

3.3.2 Eg. MySQL

CHAPTER 4

SYSTEM DESIGN

4.1 INTRODUCTION

4.2 UML DIAGRAM

4.2.1 USE CASE DIAGRAM

Explanation, Diagram

4.2.1 SEQUENCE DIAGRAM

Explanation, Diagram

4.2.2 State Chart Diagram

Explanation, Diagram

4.2.2 Activity Diagram

Explanation, Diagram

4.2.3 Class Diagram

Explanation, Diagram

4.2.4 Object Diagram

Explanation, Diagram

4.2.5 Component Diagram

Explanation, Diagram

4.2.8 Deployment Diagram

Explanation, Diagram

4.2.9 Collaboration Diagram

Explanation, Diagram

4.3 USER INTERFACE DESIGN USING FIGMA

Form Name: abcc

Screenshot

Form Name: abcc

Screenshot

All Forms of Mini Project (minimum 8)

4.4 DATABASE DESIGN

4.4.1 Relational Database Management System (RDBMS)

4.4.2 Normalization

4.4.3 Sanitization

4.4.4 Indexing

4.5 TABLE DESIGN

1.Tbl_users_login

Primary key: **loginid**

Foreign key: **loginid** references table **Tbl_users_login**

No:	Field name	Datatype (Size)	Key Constraints	Description of the field

CHAPTER 5

SYSTEM TESTING

5.1 INTRODUCTION

Explanation

5.2 TEST PLAN

Explanation

5.2.1 Unit Testing

explanation

5.2.2 Integration Testing

Explanation

5.2.3 Validation Testing or System Testing

Explanation

5.2.4 Output Testing or User Acceptance Testing

explanation.

5.2.5 Automation Testing

explanation.

5.2.6 Selenium Testing

explanation.

Example:**Test Case 1****Code**

```
package test1;
import org.openqa.selenium.By;
public class login {
    public static void main(String[] args) {
        System.setProperty("webdriver.chrome.driver", "C:\\\\Users\\Sangeetha\\Down.
        WebDriver driver=new ChromeDriver();
        driver.get("http://localhost/Crime-Reporting-System_user/userlogin.php");
        driver.findElement(By.id("exampleInputEmail1")).sendKeys("chithra@gmail.com");
        driver.findElement(By.id("exampleInputPassword1")).sendKeys("Chithra@12");
        driver.findElement(By.id("login")).click();
        String actualUrl="http://localhost/Crime-Reporting-System_user/userhome.php";
        String expectedUrl= driver.getCurrentUrl();
        if(actualUrl.equalsIgnoreCase(expectedUrl)) {
            System.out.println("Test passed");
        } else {
            System.out.println("Test failed");
        }
    }
}
```

Eg.Screenshot

```
Starting ChromeDriver 103.0.5060.53 (a1711811edd74ff1cf2150f36ffa3b0dae40b17f-refs/branch-heads/50600{#f
Only local connections are allowed.
Please see https://chromedriver.chromium.org/security-considerations for suggestions on keeping ChromeDr
ChromeDriver was started successfully.
Jul 21, 2022 3:08:40 PM org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected upstream dialect: W3C
Jul 21, 2022 3:08:40 PM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
INFO: Found exact CDP implementation for version 103
Test passed
```

Eg. Test Report

Test Case 1					
Project Name:					
Login Test Case					
Test Case ID: Test 1			Test Designed By:		
Test Priority(Low/Medium/High):			Test Designed Date:		
Module Name:			Test Executed By :		
Test Title :			Test Execution Date:		
Description:					
Pre-Condition : User has valid username and password					
Step	Test Step	Test Data	Expected Result	Actual Result	Status(Pass/ Fail)
1					
2					
3					
4					
5					
6					
7					
Post-Condition:					

Test Case 2:

Code

Screenshot

Test report

Minimum 4 test cases (1 login 3 functionalities)

CHAPTER 6

IMPLEMENTATION

6.1 INTRODUCTION

Explanation

6.2 IMPLEMENTATION PROCEDURES

Explanation

6.2.1 User Training

Explanation

6.2.2 Training on the Application Software

Explanation

6.2.3 System Maintenance

Explanation

6.2.4 Hosting

Explanation

Eg.000Webhost

Explanation

Procedure for hosting a website on 000Webhost:

Step 1: explanation

Step 2: explanation

Step 3: explanation

.

.

Hosted Website:

Hosted Link: <https://abc.000webhostapp.com>

Hosted Link QR Code

Screenshot

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

.

7.2 FUTURE SCOPE

.

CHAPTER 8

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WEBSITES:

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CHAPTER 9

APPENDIX

9.1 Sample Code

Main functionalities

9.2 Screen Shots

9.3 Git Log

